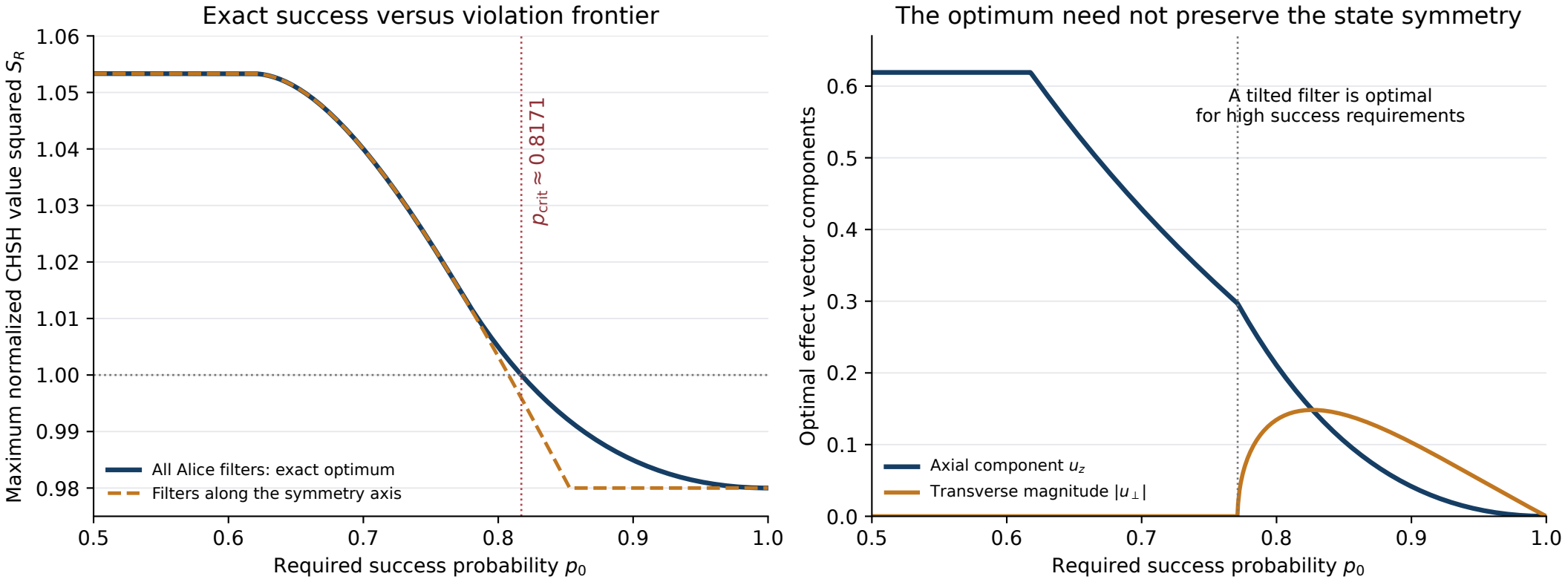
Exact success versus violation frontier
Maximum normalized CHSH value squared S_R
1.06
1.05
1.04
1.03
1.02
1.01
1.00
0.99
0.98
$p_{\mathrm{crit}} \approx 0.8171$
All Alice filters: exact optimum
Filters along the symmetry axis
0.5
0.6
0.7
0.8
0.9
1.0
Required success probability p_0
The optimum need not preserve the state symmetry
Optimal effect vector components
0.6
0.5
0.4
0.3
0.2
0.1
0.0
A tilted filter is optimal
for high success requirements
Axial component u_z
Transverse magnitude $|u_\perp|$
0.5
0.6
0.7
0.8
0.9
1.0
Required success probability p_0